GitHub Pages Automatic Deployment Experiment Report

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Student ID: (Fill in your student ID)
Experiment Date: (Fill in the completion date)

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1. Experiment Objective

The purpose of this experiment is to learn how to deploy a static website using GitHub Pages and GitHub Actions. The specific goals include:

1. Configure the GitHub Pages service
2. Use GitHub Actions to implement automatic website deployment
3. Verify the live status of the website and record the complete experiment process

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2. Experimental Environment

- Operating System: Windows 11
- Web Browser: Google Chrome
- Platform: GitHub official website
- Experiment Repository: Cloud-Computing

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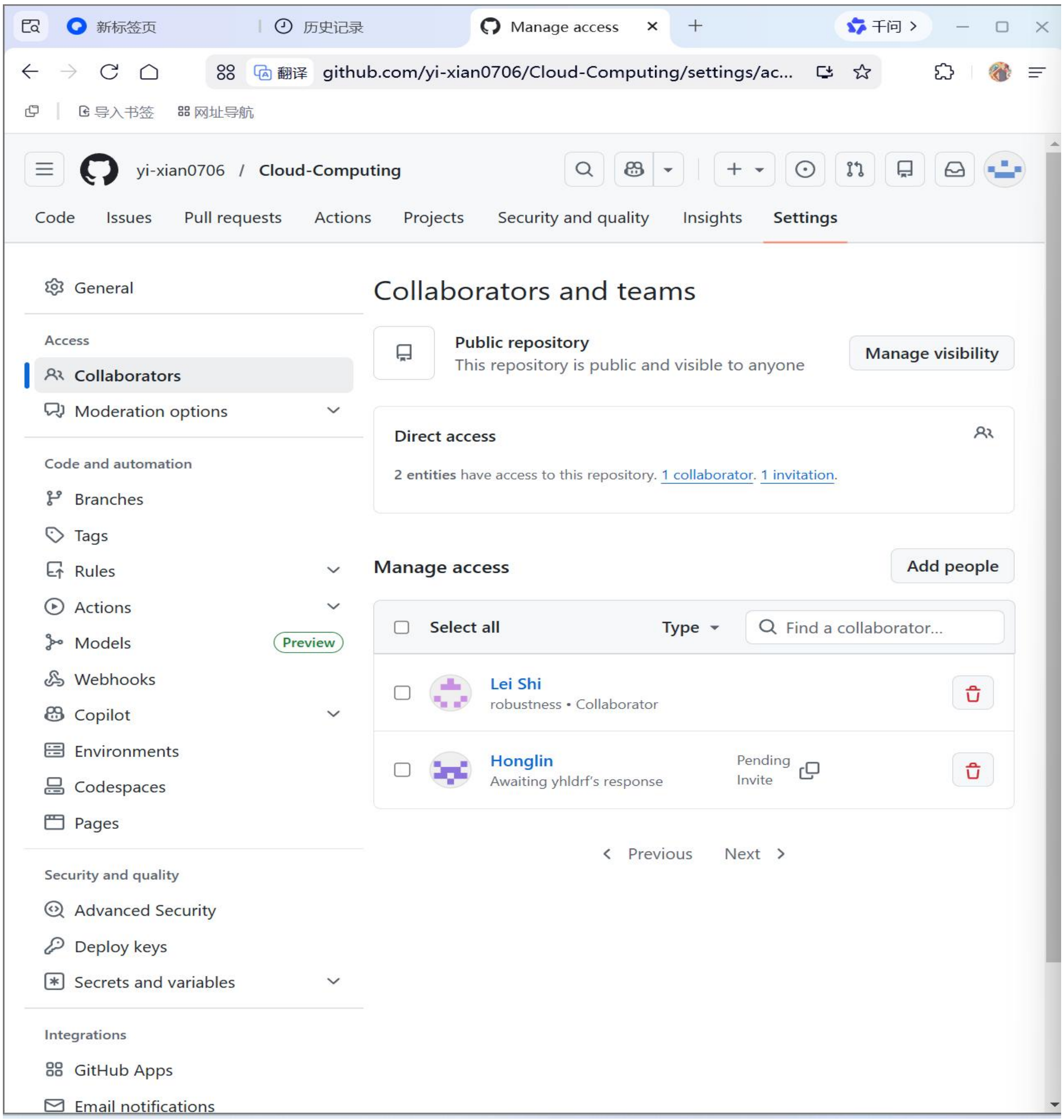
3. Experimental Steps and Screenshot Instructions

Step 1: Prepare a Public Repository

1. Log in to your GitHub account and open the Cloud-Computing repository page.
2. Go to the settings page, find the danger zone in the general settings, and change the repository visibility from private to public.
3. Confirm the change and refresh the page. The repository status will be updated to public.

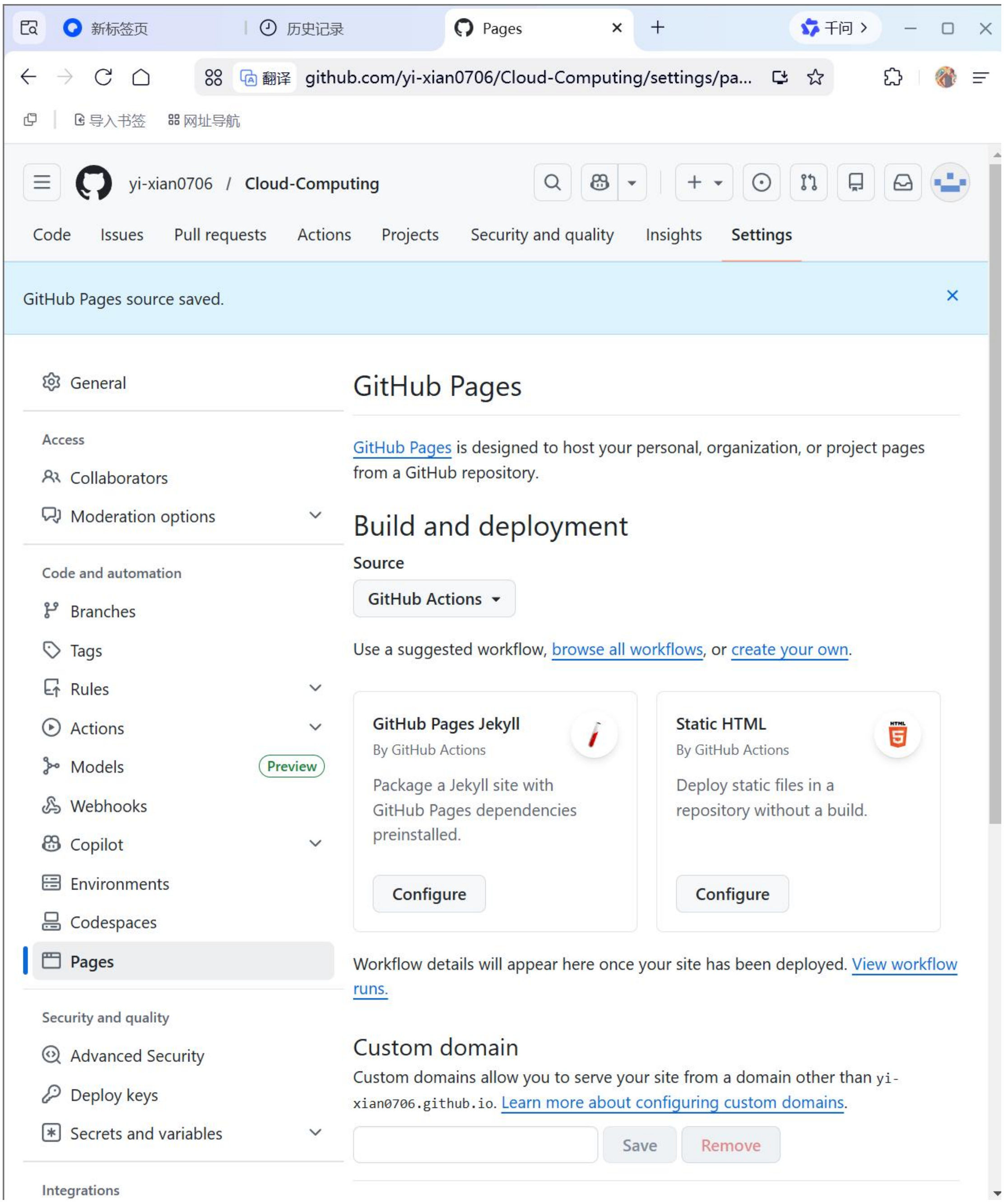


Step 2: Configure the GitHub Pages Service

1. On the repository settings page, find and enter the Pages settings option.

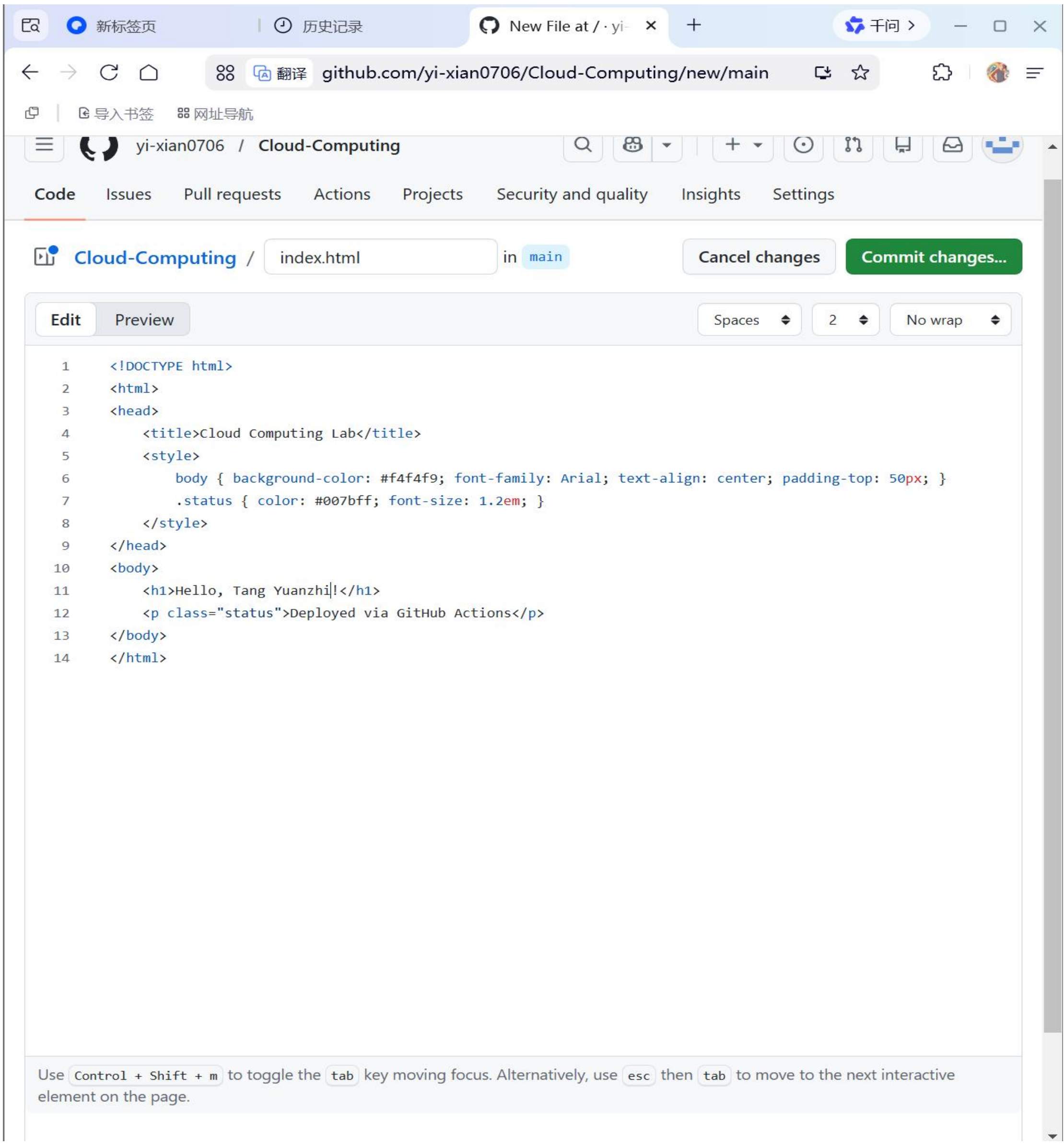
2. In the build and deployment section, set the deployment source to GitHub Actions.

3. Select the static HTML workflow template, generate the configuration file, and submit and save it.



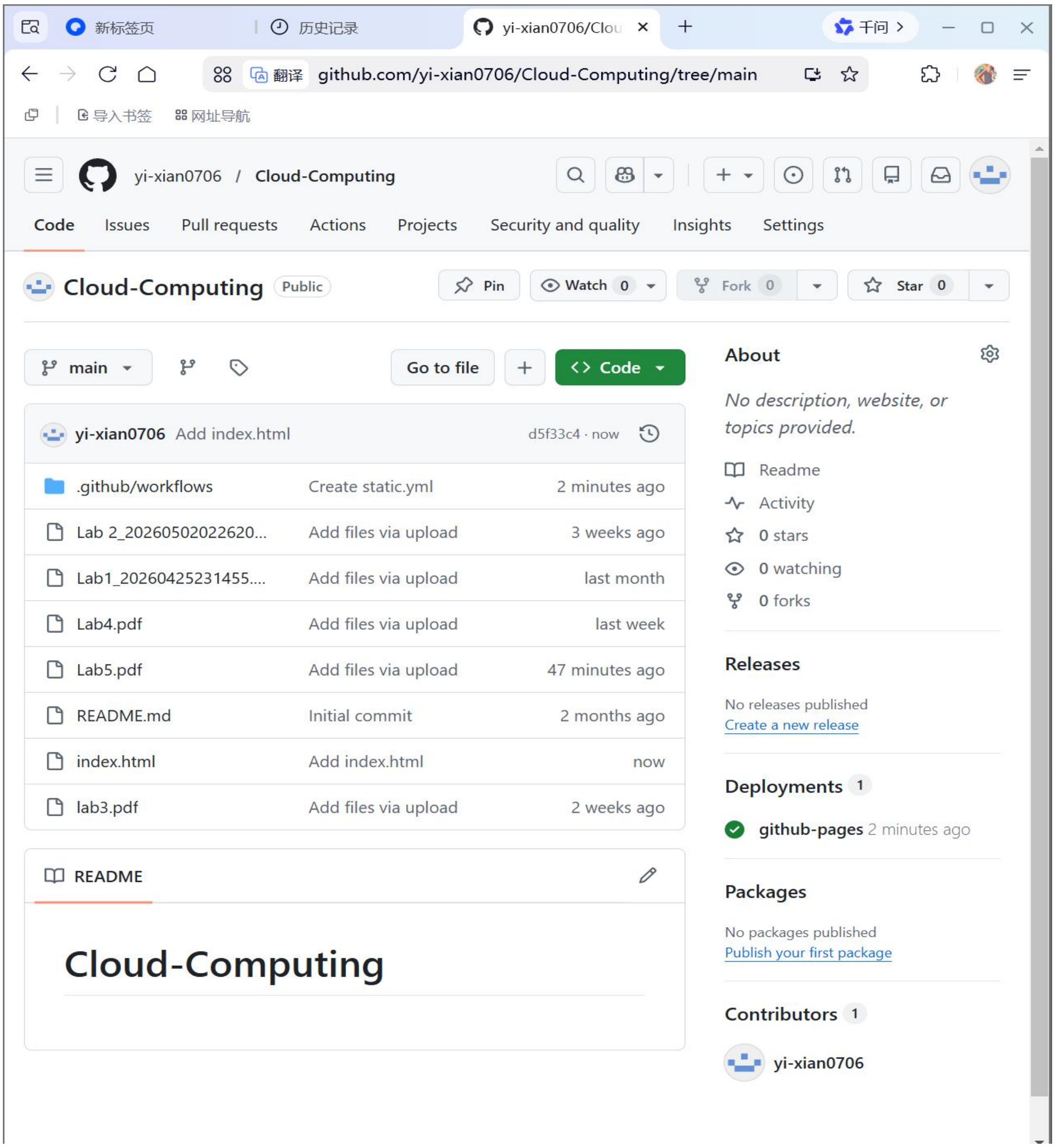
Step 3: Create the Web Page File

1. Return to the repository homepage, click add file, and select create new file.
2. Name the file index.html and enter the web page content in the editing area.
3. Submit the file with a brief message and complete the creation.



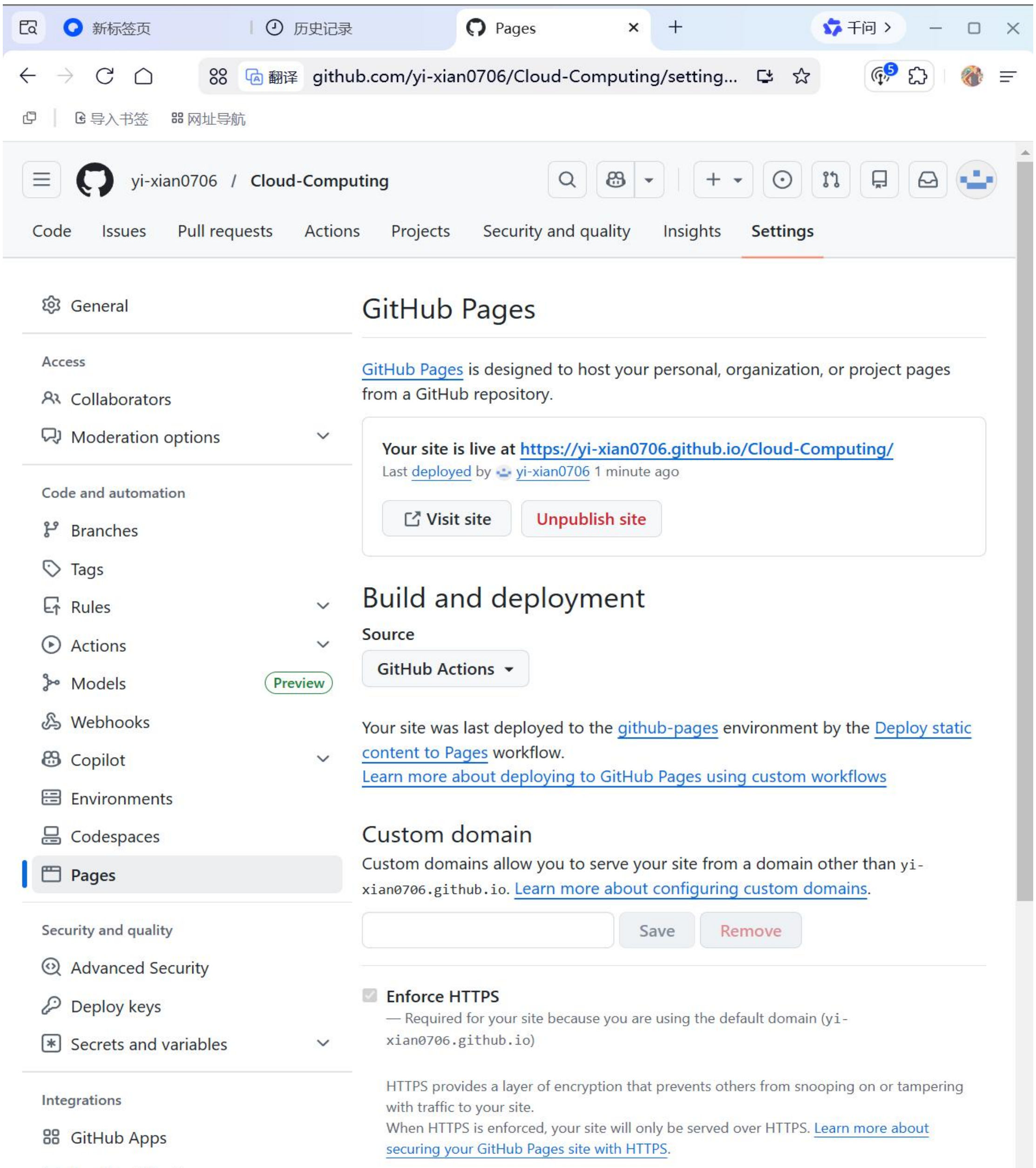
Step 4: Monitor the Deployment Workflow

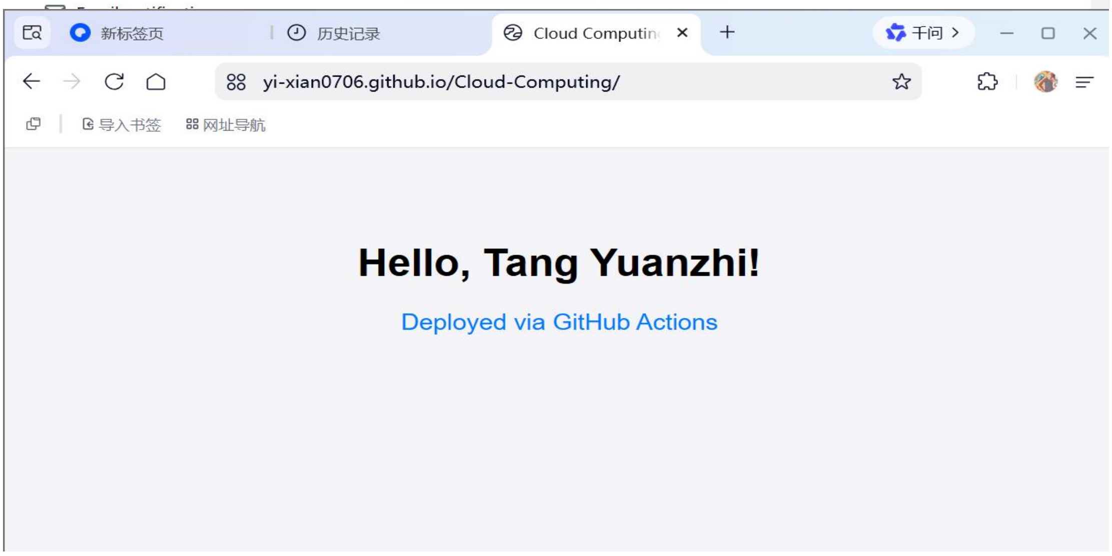
1. Click the Actions tab in the repository.
2. Observe the running status of the pages-build-deployment workflow.
3. Wait for the workflow to complete, which is indicated by a green check mark.



Step 5: Verify the Live Website

1. Return to the Pages settings page.
2. Copy the live site URL displayed on the page.
3. Open the URL in a new browser tab to confirm the website loads correctly.


新标签页
历史记录
Pages
千问
github.com/yi-xian0706/Cloud-Computing/setting...
导入书签
网址导航
yi-xian0706 / Cloud-Computing
Code
Issues
Pull requests
Actions
Projects
Security and quality
Insights
Settings
General
Access
Collaborators
Moderation options
Code and automation
Branches
Tags
Rules
Actions
Models
Preview
Webhooks
Copilot
Environments
Codespaces
Pages
Security and quality
Advanced Security
Deploy keys
Secrets and variables
Integrations
GitHub Apps
GitHub Pages
GitHub Pages is designed to host your personal, organization, or project pages from a GitHub repository.
Your site is live at https://yi-xian0706.github.io/Cloud-Computing/
Last deployed by yi-xian0706 1 minute ago
Visit site
Unpublish site
Build and deployment
Source
GitHub Actions
Your site was last deployed to the github-pages environment by the Deploy static content to Pages workflow.
Learn more about deploying to GitHub Pages using custom workflows
Custom domain
Custom domains allow you to serve your site from a domain other than yi-xian0706.github.io. Learn more about configuring custom domains.
Save
Remove
Enforce HTTPS
— Required for your site because you are using the default domain (yi-xian0706.github.io)
HTTPS provides a layer of encryption that prevents others from snooping on or tampering with traffic to your site.
When HTTPS is enforced, your site will only be served over HTTPS. Learn more about securing your GitHub Pages site with HTTPS.


新标签页
历史记录
Cloud Computin
千问
yi-xian0706.github.io/Cloud-Computing/
导入书签
网址导航
Hello, Tang Yuanzhi!
Deployed via GitHub Actions

4. Experimental Results

The experiment was completed successfully. The static website is now live and accessible through the generated URL. Any changes made to the web page file will trigger an automatic redeployment via GitHub Actions, confirming that the entire automation process works as intended.

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5. Problems Encountered and Solutions

1. **Problem**: GitHub Pages was unavailable because the repository was private.
Solution: Changed the repository visibility to public in the settings.

2. **Problem**: The deployment workflow did not start after setting the source to GitHub Actions.
Solution: Selected the static HTML workflow template to generate the required configuration file, which triggered the pipeline.

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6. Conclusion

This experiment demonstrated the complete process of cloud-based static website deployment using GitHub Pages and GitHub Actions. The key takeaways include that GitHub Pages provides free hosting for static content, GitHub Actions automates the deployment process, and updates to the web page can be automatically deployed without manual intervention. The experiment verified the feasibility and convenience of cloud-based deployment tools.